Short questions

|!|BCESA0100000001|!| ★★ **Cross topic**

Figure 1 and Figure 2 below show root hairs on the root surface. The two images were taken using different types of microscopes.

Figure 1



Figure 2



(a) Which of the two images was taken using a scanning electron microscope? Justify your answer. (2 marks)

(b) Describe and explain ***two*** adaptations of the root observable in the figures for water absorption. (4 marks)

##

(a) Figure 1 (1)
The surface of the root is viewed in three-dimensional perspective. (1)

(b) The root has numerous root hairs (1)
to increase the surface area for the absorption of water. (1)
Root hairs are fine and long. (1)
They can reach the spaces between soil particles to absorb water. (1)

(6 marks)

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##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Applying concepts	4

|!|BCESB0100000002|!| ★

For each of the elements listed in column I, select from column II one phrase that is related to it. Put the letter in the space provided. Some may require more than one answer. (6 marks)

Column I		***Column II***
Magnesium	(a)_____	A. yellow leaves when in deficient
Nitrogen	(b)_____	B. essential for the formation of nucleic acids
Phosphorus	(c)_____	C. essential for the formation of chlorophyll
		D. essential for the transport of nutrients
		E. slow growth of fruits when in deficient

##

(a) A, C (1, 1)

(b) A, B, C (1, 1, 1)

(c) B (1)

(6 marks)

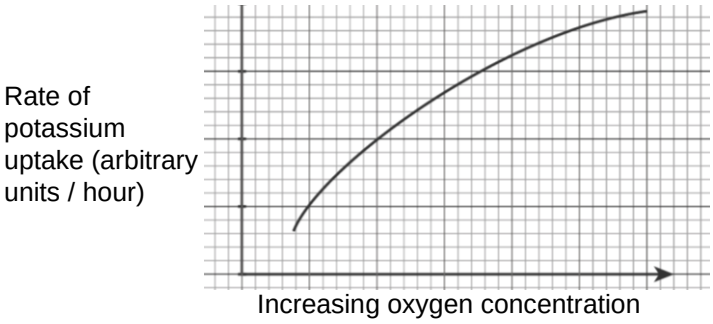
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	1

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|!|BCESB0100000003|!| ★

The graph below shows the effect of change in the oxygen concentration on the absorption rate of potassium ions from soil water by the roots of a seedling.



Describe and explain the effect of increased oxygen concentration on the absorption of potassium ions from soil water. (3 marks)

##

The rate of absorption of potassium ions from soil water increases with the increased oxygen concentration. (1)

With more oxygen, the root cells carry out respiration faster, releasing more energy (1)

for the uptake of potassium ions by active transport. (1)

(3 marks)

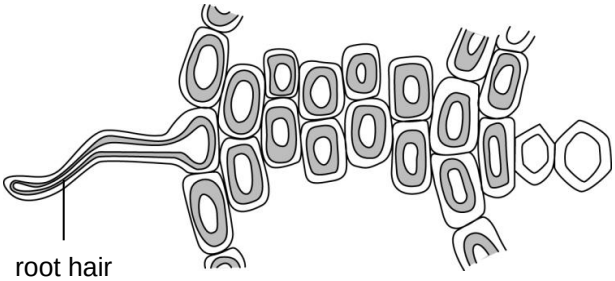
##

Skills assessed	Marks
Analysing data	1
Applying concepts	2

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|!|BCESB0100000004|!| ★

The diagram below shows the transverse section of a certain part of a young root in the soil.



(a) With reference to the diagram, explain how the root hair cells are adapted for absorption of water and mineral ions. (4 marks)

(b) Explain how water moves from the soil into the root hair cell. (3 marks)

##

(a) Root hairs are fine and long. (1)
They can reach the spaces between soil particles to absorb water and minerals. (1)
No cuticle is present in the root hair cells. (1)
This allows water and minerals to enter the root hair cells easily. (1)

(b) Since the cytoplasm and the cell sap of root hair cells contain dissolved substances, (1)
the water potential of root hair cells is usually lower than that of soil water. (1)
As a result, water is absorbed into the root hair cells by osmosis. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Understanding basic concepts	2
(b)	Understanding basic concepts	3

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|!|BCESC0100000005|!|

The paragraph below describes how water in the root hair cells enters the xylem of the root. Fill in the blanks with suitable words. (4 marks)

Due to the absorption of water, the water potential of the root hair cells is (a) _____ than the neighbouring cells in the (b) _____. Water moves inwards from cell to cell by (c) _____. Meanwhile, some water moves between the (d) _____ of adjacent cells without entering the cytoplasm or the vacuole of the cells.

##

(a) higher (1)
(b) cortex (1)
(c) osmosis (1)
(d) cell wall (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESB0100000006|!| ★

In the table below, indicate, by putting a tick (✓) in the appropriate box, the direction of movement of the gases that you would expect on a bright, dry day. Give a reason for each of your answers. (Note: Assume the stomata are partially open.) (6 marks)

	Net movement of gas / water vapour			**Reason**
	into leaf	**out of leaf**	**zero**	
(a)Oxygen				
(b)Carbon dioxide				
(c)Water vapour				

##

(a) Out of leaf (1)
Reason: Rate of photosynthesis is faster than that of respiration (1)

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(b) Into leaf (1)

Reason: Carbon dioxide is the raw material for photosynthesis (1)

(c) Out of leaf (1)

Water vapour diffuses down the concentration gradient (1)

(6 marks)

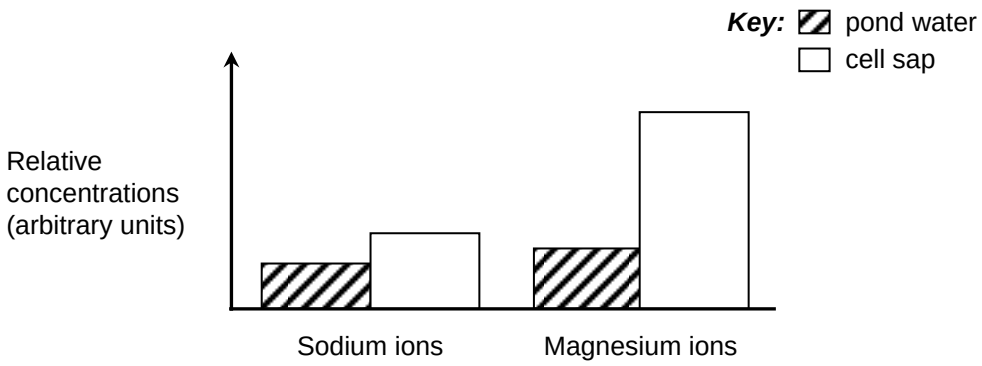
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCESA0100000007|!| ★★

Bar chart I shows the relative concentrations of sodium and magnesium ions in pond water in the cell sap of a green alga. The data were collected at 20 ^{o}C under aerobic conditions.

Bar chart I



(a) Give evidence from bar chart I to support the following claims:

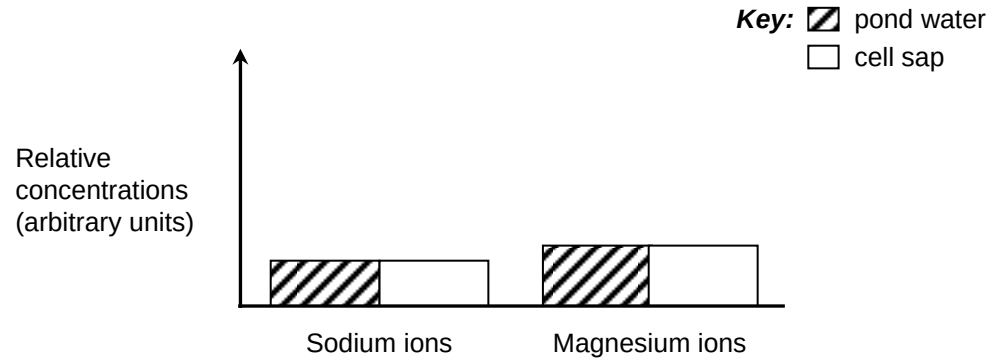
(i) Ion uptake is carried out by active transport. (1 mark)

(ii) Ion uptake is selective. (1 mark)

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The experiment was repeated at 20 ^{o}C under anaerobic conditions. The results obtained are shown in bar chart II.

Bar chart II



(b) Account for the differences in ion uptake shown in bar charts I and II. (3 marks)

(c) Suggest ***one*** importance of magnesium to the survival of the green algal cells. (2 marks)

##

(a) (i) The concentrations of sodium and magnesium ions inside the cell sap are higher than those in the pond water. / The uptake of ions is against the concentration gradient. (1)

(ii) The uptake of magnesium ions is greater than that of sodium ions. (1)

(b) The uptake of both sodium and magnesium ions stops under anaerobic conditions. (1)

The uptake process is carried out by active transport. (1)

Under anaerobic conditions, less energy / ATP is produced. (1)

(c) For synthesis of chlorophyll, (1)

which carries out photosynthesis to produce food. (1)

(7 marks)

##

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	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)	Applying concepts	3
(c)	Understanding basic concepts	2

|!|BCESB0100000008|!| ★

The table below shows the concentrations of some ions in the cells of an alga, and in the pond water in which this alga lives.

	Ion concentration (mg / L)			
	Calcium	**Chloride**	**Magnesium**	**Potassium**
Cell sap	350	3710	270	2200
Pond water	25	34	37	2
Accumulation ratio	14 : 1	109 : 1	7.3 : 1	X

(a) Calculate the accumulation ratio for potassium (X). Show your working. (2 marks)

(b) Compare the ion concentrations in the cell sap with those in the pond water. State ***one*** general trend observed. (1 mark)

(c) Do the data in the table support the following claims? Explain your answer.

(i) The cell membrane is differentially permeable. (2 marks)

(ii) Ion uptake occurs as a result of diffusion. (2 marks)

##

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(a) 2200 / 2 (1)
= 1100 : 1 (1)

(b) Ion concentration is significantly higher in the cell sap than in the pond water. (1)

(c) (i) Yes. (1)
In the cell sap, the concentrations of calcium ions and chloride ions are different. (1)

(ii) No. (1)
The uptake of ions by the alga is against the concentration gradients / from low concentrations to high concentrations. (1)

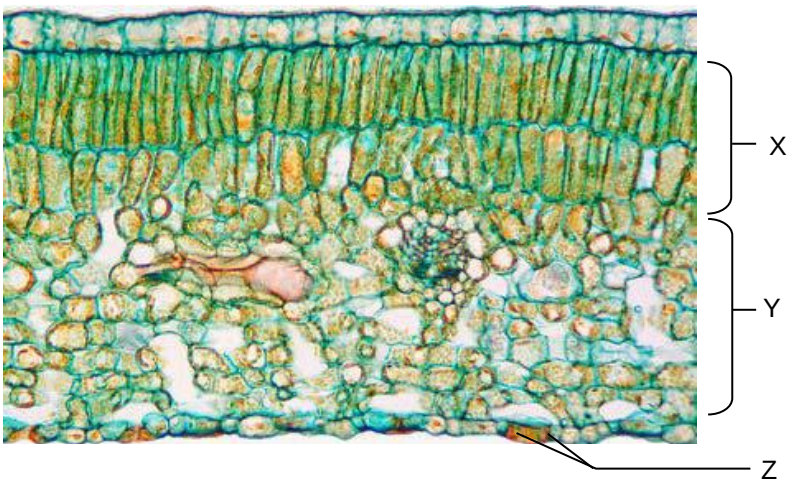
(7 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Analysing data	1
(c)(i)	Analysing data	2
(c)(ii)	Analysing data	2

|!|BCESA0100000009|!| ★★

A student examined a leaf with a light microscope. The following figure shows the magnified image of the cross section of the leaf, with a magnification of ×100.



(a) If the student used a 20X eyepiece in the examination, what would be the magnification of the objective? (1 mark)

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(b) (i) State how tissue X is adapted for carrying out photosynthesis. (1 mark)

(ii) State how tissue Y is adapted for carrying out gas exchange. (1 mark)

(c) (i) In the space below, sketch the appearance of cells Z in the dark. (2 marks)

(ii) What is the significance of the appearance of cells Z stated in (c)(i) to the plants? (1 mark)

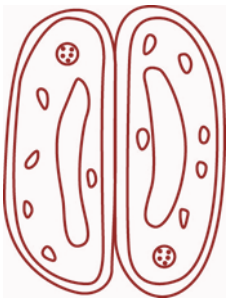
##

(a) 5X objective (1)

(b) (i) Tissue X is near the epidermis / contains cells that have many chloroplasts. (1)

(ii) Tissue Y contains cells that are loosely packed with many air spaces among them. (1)

(c) (i)



Guard cells in the dark condition

- Drawing of two sausage-shaped guard cells (1)
- Drawing of closure of stoma (1)

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(ii) To prevent excessive water loss through stomata. (1)

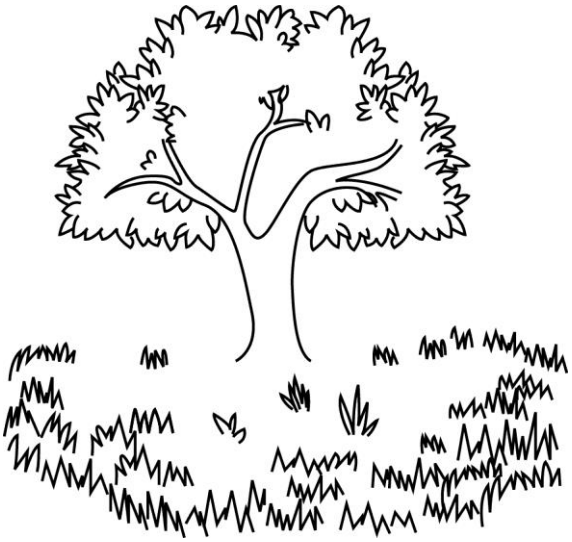
(6 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)(i)	Presenting information	2
(c)(ii)	Understanding basic concepts	1

|!|BCESA0100000010|!| ★★ **SI**

Jenny found that if grass grows closer to a tree, the grass will grow less well. The following diagram shows what she observed.

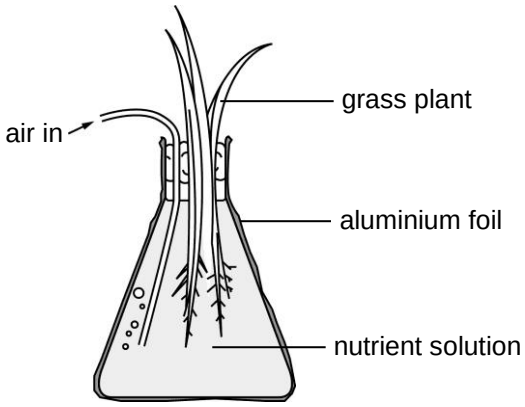


Based on this observation, Jenny hypothesized that since most nitrates in the soil are taken up by the tree, the grass besides the tree cannot get sufficient nitrates for growth.

(a) Suggest ***another*** hypothesis for the phenomenon as shown above. (1 mark)

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(b) Jenny carried out an experiment to test her hypothesis. She cultured grass samples with nutrient solutions of different nitrate concentrations. The following shows the experimental set-ups.



(i) State the dependent variable of this investigation. (1 mark)

_____

(ii) State ***two*** variables that need to be controlled in this experiment. (2 marks)

_____

_____

(iii) State ***one*** way to increase the accuracy of the results. (1 mark)

_____

_____

##

(a) Grass that grows near the tree cannot get enough light for carrying out photosynthesis. (Or other reasonable answer) (1)

(b) (i) Height of the grass / dry mass of the grass (Or other reasonable answer) (1)

(ii) Any ***two*** of the following: (2)

- temperature (1)
- carbon dioxide level (1)
- light intensity (1)
- other mineral levels (1)
- density of grass plants per set-up (1)

(Or other acceptable answers)

(iii) Any ***one*** of the following: (1)

- repeat the experiment to obtain more data (1)
- use nutrient solutions with a wider range of nitrate concentrations (1)

(Or other acceptable answers)

(5 marks)

##

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	Skills assessed	Marks
(a)	Practical skills	1
(b)(i)	Practical skills	1
(b)(ii)	Practical skills	2
(b)(iii)	Practical skills	1

|!|BCESC0100000011|!|

Nitrogen, phosphorus and magnesium are elements essential for plant's growth.

State ***one*** function of the following elements and ***one*** corresponding deficiency symptom in plants.

(a) nitrogen (2 marks)

(b) phosphorus (2 marks)

(c) magnesium (2 marks)

##

(a) Nitrogen is used for the synthesis of proteins / nucleic acids / chlorophyll. (1)
Nitrogen deficiency will lead to poor growth and yellow leaves. (1)

(b) Phosphorus is a component of nucleic acid / cell membrane / enzyme. (1)
Phosphorus deficiency will lead to poor growth and lead to the formation of purple patches on leaves. (1)

(c) Magnesium is required for chlorophyll formation. (1)
Magnesium deficiency will lead to poor growth and yellow leaves. (1)

(6 marks)

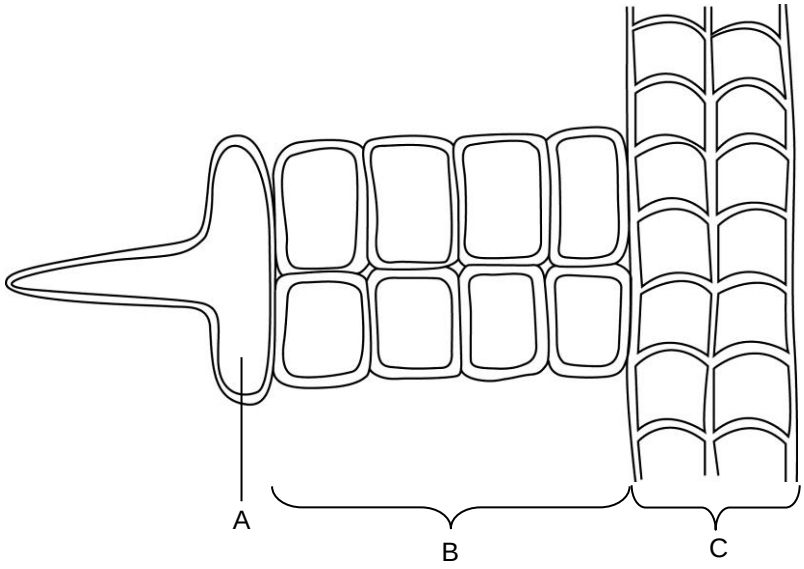
##

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	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCESB0100000012|!| ★

The diagram below shows the different cells found in a root.



(a) State and explain ***one*** adaptation of cell A to perform its function. (2 marks)

(b) Describe ***two*** ways by which water is transported in tissue B. (2 marks)

(c) What causes the water to be drawn up through tissue C to the leaf? (1 mark)

##

(a) Cell A is not covered by cuticle, (1)
hence water can enter easily. (1)
OR
Cell A develops root hairs (1)

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which greatly increase the surface area of roots for water and mineral uptake. (1)

(b) Water is transported across the cortex (tissue B) from cell to cell by osmosis. (1)

Water is transported along the cell walls of the adjacent cortex cells. (1)

(c) Transpiration pull (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1

|!|BCESC0100000013|!|

The following paragraph summarizes the way of absorption of minerals by roots. Complete the paragraph with suitable word(s). (4 marks)

The concentration of mineral ions in the soil water is usually (a)_____ than that in the root cells. Most mineral ions are thus absorbed into the root cells against the (b)_____ by (c)_____. This requires (d)_____ released from respiration.

##

(a) lower (1)

(b) concentration gradient (1)

(c) active transport (1)

(d) energy (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESB0100000014|!| ★

Marcus collected samples of ocean water at different depths at intervals and measured the changes in oxygen content of the water within a certain period of time. From the measurement, he could calculate the rate of oxygen production and consumption by the algae in water. The following table shows the results.

Water depth (m)	**Rate of oxygen production ($\mu mol\ O_2\ dm^{-3}\ day^{-1}$)**	**Rate of oxygen consumption ($\mu mol\ O_2\ dm^{-3}\ day^{-1}$)**
2	6.8	3
9	6.6	3.9
15	4.2	2.2
21	3.6	2.8
26	3.3	3.7
30	1.5	2.1
46	1.2	1.2
58	1	1.2

(a) According to the table, describe and explain the relationship between the depth of water and the rate of oxygen production. (3 marks)

(b) The rate of net oxygen production can be calculated by:

Rate of net oxygen production = Rate of oxygen production – Rate of oxygen consumption

According to the equation shown above, plot a graph to show the relationship between the depth of water and the rate of net oxygen production by the algae in the graph paper below. (4 marks)

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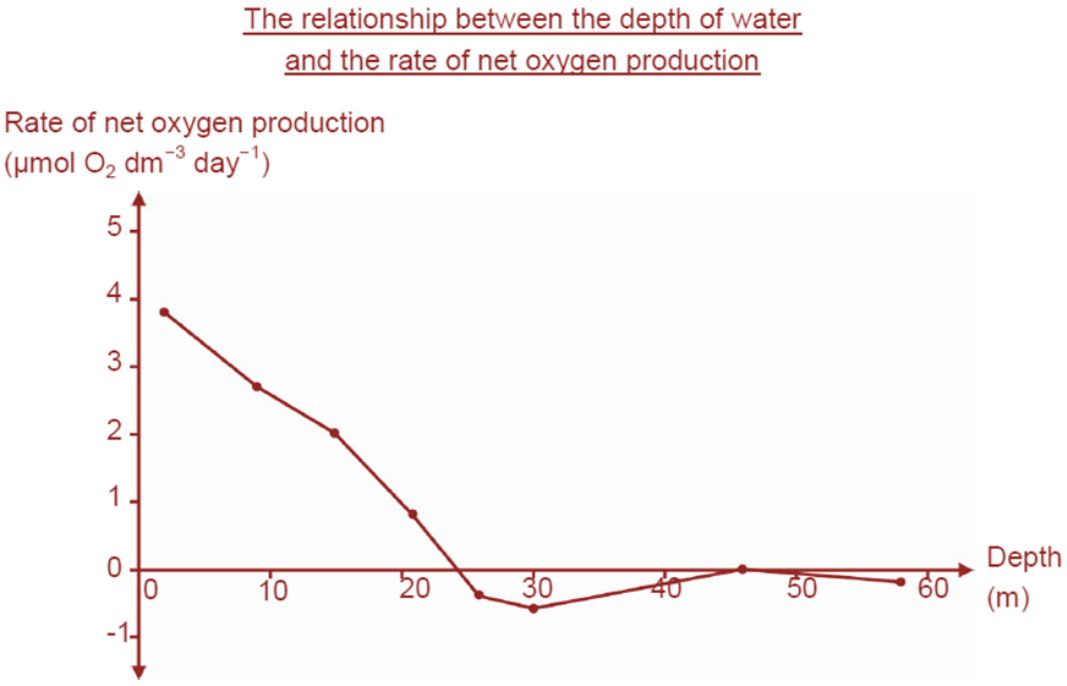
##

(a) The greater the water depth, the lower the rate of oxygen production. (1)

As there is lower light intensity at greater water depth, (1)

less light is absorbed by the algae and so the rate of photosynthesis decreases. (1)

(b)



- Correct labelling of axis (1)
- Correct plotting of points (2)
- Correct title (1)

(7 marks)

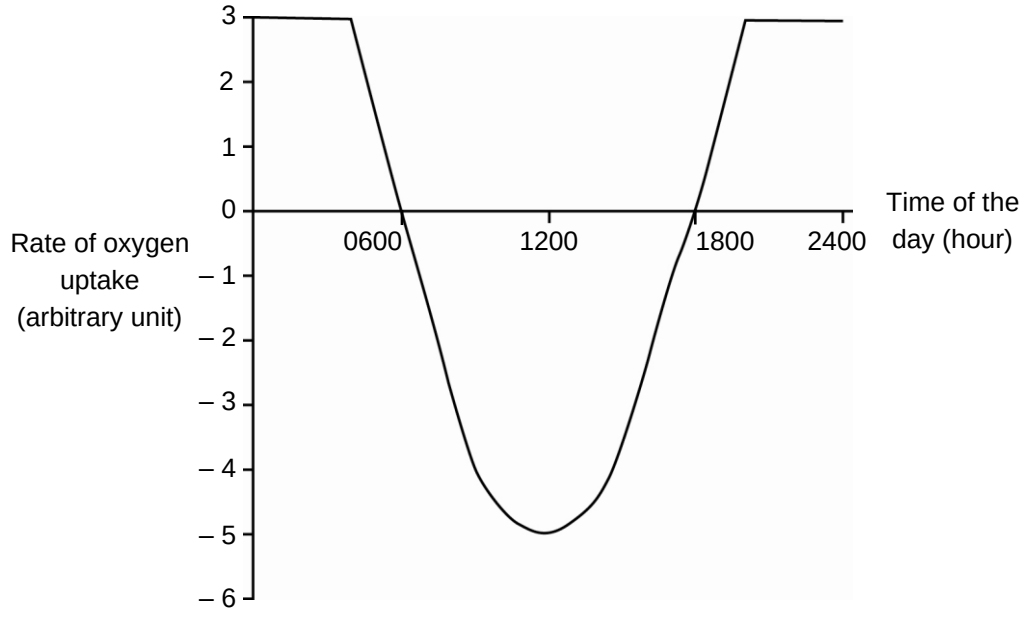
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##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Presenting information	4

|!|BCESB0100000015|!| ★

The following graph shows the rate of oxygen uptake by the water lily during a day.



(a) State the time at which the water lily carries out photosynthesis at the maximum rate. (1 mark)

(b) What is the implication of having a negative uptake of oxygen for the water lily? (1 mark)

(c) According to the graph, how does the water lily affects the oxygen content in water during night time? (2 marks)

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##

(a) 12:00 (1)

(b) A negative uptake of oxygen implies that the photosynthetic rate of the water lily is higher than the rate of respiration. (1)

(c) During night time, there is a net uptake of oxygen by the water lily. (1)
Therefore, the oxygen content in water decreases. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	1
(c)	Analysing data	2

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